PROPOSAL · OVERALL

Agentic Dataset Context Engine

A knowledge-graph-backed, human-supervised agent system that relates datasets, builds a living data dictionary, and reconstructs how reports are produced.

Document	Overall product and architecture proposal (revised with design decisions)
Prepared for	Data & Engineering Leadership
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Companion	Starting-stage proposal: Qlik-to-SQL reconstruction

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1. Executive summary

Organizations hold hundreds of datasets whose meaning, keys, relationships and origins are poorly documented, and many reports whose logic is buried in BI applications. Every integration, migration, audit or reconciliation starts by rediscovering the same facts, and that knowledge is lost when the project ends.

We propose an **agentic dataset context engine**: specialist AI agents that share one continuously growing **knowledge graph**. They read the data estate (Dremio), BI applications (Qlik), schemas and documents; profile datasets; discover keys and joins; detect overlapping and duplicate datasets and recommend golden records; draft definitions and lineage; publish a data dictionary; and reconstruct the SQL behind reports. Statistics answer what they can and AI models handle judgment. Every result carries evidence and passes human review before it is trusted.

The value compounds: each dataset, document, report and human correction makes the next analysis faster and more accurate. Delivery starts with the capability that proves the central claim, **reconstructing Qlik report logic as validated Dremio SQL**, and builds the wider platform on the same foundations.

> **Ask: approve the programme as described, starting with the 12-week Qlik-to-SQL stage, with the wider platform building on the same foundations from week 6.**

2. The problem

- **Knowledge does not compound.** Schema matches, keys and join logic are re-derived every time a dataset is used, with no record of what was learned before.
- **The data estate has grown without structure.** Many views overlap, data is copied, and it is unclear which dataset is the authoritative (golden) record and which are dumps or snapshots.
- **Reports cannot be traced.** Report logic lives in BI load scripts, intermediate files and chart expressions. Reconciling a published figure with its source is manual detective work.
- **Documentation is missing or scattered** across people, spreadsheets and old documents, and arrives late, disconnected from earlier analysis.
- **Regulatory and migration pressure.** Lineage for reported figures must be shown, and legacy BI applications must be understood before they are migrated or retired.

3. Proposed solution

Inputs

Input	Examples	What the system does with it
Data estate	Dremio tables and views, query history; uploaded files	Profiles, finds keys, relates datasets, detects duplicates and golden records
BI applications	Qlik load scripts, project folders, reload logs, QVD files, exported reports	Extracts report logic and lineage; reconstructs it as SQL and validates it
Context	DDL, view definitions, data dictionaries, wikis, business documents	Extracts documented facts and links them to tables and columns
Scope tags	"Belongs to source Sales ERP"	Attaches new material to a known source so it inherits existing knowledge

Outputs

Output	Format
Dataset profiles, semantic types, sensitivity tags	Workbench, JSON
Key and join recommendations with evidence	Workbench, graph
Estate analysis: duplicate clusters, recommended golden records, stale datasets	Report, graph
Data dictionary and business glossary	Excel, Markdown, PDF, dbt YAML
Lineage map	Graph view, OpenLineage
Reconstructed report SQL with validation level	SQL, migration pack, report
Knowledge graph access for other tools and AI agents	MCP server, API

4. Why it is product-worthy

Reason	Detail
Expensive, recurring problem	Documenting an undocumented database takes weeks of expert effort; one study estimates USD 12,000–48,000 per moderately complex database [1]. It recurs in every migration, acquisition, audit and regulatory review.
Value compounds (the moat)	Every confirmed key, definition, join and report mapping becomes a trusted fact reused later. After months of use, the knowledge graph is an organization-specific asset that cannot be copied.
Fills gaps catalogs leave open	Works where documentation and code are missing; relates disconnected sources; reconstructs and validates report logic; handles messy estates with duplicates; human-supervised with a full audit trail.
Fits existing stacks	Feeds the catalogs organizations already own (OpenLineage, dbt, MCP) instead of replacing them, and deploys inside the client's environment.
Clear buyers	Regulatory reporting lineage, BI and warehouse migration, mergers and acquisitions, audit and finance, catalog enrichment.

Category	Examples	Where we differ
Enterprise data catalogs	Atlan, Collibra, Alation, Informatica	They rely on connectors, query logs and existing code; we infer from data and BI logic when documentation is missing, and feed their catalogs
Platform-native governance	Unity Catalog, Snowflake Horizon, Purview, Dataplex	Limited to one platform; little cross-source reasoning or report reconstruction
Migration and conversion tools	Commercial BI converters	Target a specific destination; not connected to a knowledge base; we validate against independent evidence
Research systems	DBAutoDoc [1], LLM-FK [2], Magneto [3]	Strong single-database methods we build on; no compounding, multi-source product workflow

5. Design principles

1. **Statistics first, AI second.** Deterministic checks generate and verify candidates; AI models adjudicate and explain. Published research shows this combination clearly beats AI alone [1][2][4].
2. **Evidence for everything.** Every fact carries its source, evidence, trust level and history. Nothing is overwritten; superseded facts are kept.

3 **Humans decide what is trusted.** High-impact facts (keys, golden records, reconstructed SQL) require human confirmation, and corrections become reusable examples.

4 **Read-only towards client systems.** The system never modifies Dremio or BI applications; it recommends.

5 **Honest results.** Reconstructions report the level of evidence they reach, and "not resolved" is an allowed outcome.

6 **Built to ship.** Every external dependency sits behind an interface, so the product runs locally, in our cloud, or inside a client's environment from the same build.

6. Architecture

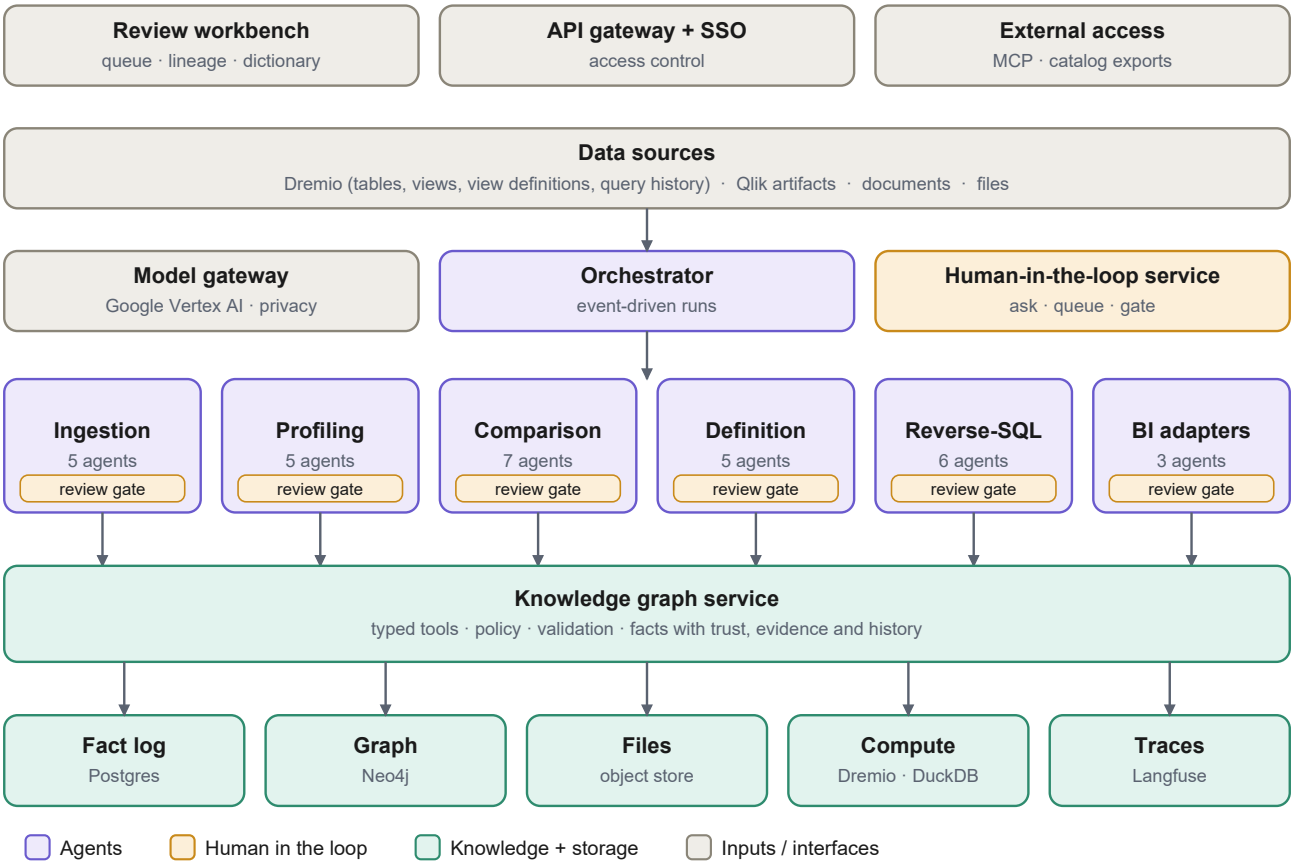


Figure 1. System architecture. Agents read from and write to one knowledge graph service, which enforces provenance, trust and access rules. Every group has a human review gate.

Component	Role
Knowledge graph service	The only way knowledge is written. About sixteen typed tools (search, context packs, lineage, join paths, propose fact, link, feedback), a policy engine, a validator that re-checks claims against the data, and context assembly within token budgets.
Knowledge core	Built on our in-house knowledge-graph engine, extended for data-catalog facts: an append-only fact log as the system of record, a graph projection for traversal, hybrid search, entity resolution and isolation that fails closed.
Orchestrator	Plans runs, reacts to change events so that only affected steps re-run, enforces budgets, and pauses for human questions.

Component	Role
Model gateway	Single route to AI models (Google Vertex AI): model routing, structured outputs, caching, budgets, response memoization for repeatable results, and the privacy boundary.
Human-in-the-loop service	Questions, review queues and approval gates, stored as data so work never blocks on a reviewer.
Review workbench	Review queue, lineage and graph explorer, dictionary view, estate analysis.

7. Agent groups

Thirty-one agents in five groups plus an orchestrator. Several are deterministic by design: statistics answer what they can, and models handle judgment.

Group	Agents	What it does
Ingestion	Intake router, schema parser, document extractor, entity linker, scope registrar	Classifies inputs, reads the Dremio catalog and view definitions, extracts documented facts, links everything to known sources
Profiling	Statistics profiler, semantic typer, sensitive-data classifier, key discoverer, domain classifier	Full-data statistics computed inside Dremio, semantic and sensitivity tags, primary keys
Comparison and estate	Candidate finder, match reranker, foreign-key detector, global verifier, insight writer, dataset equivalence detector, golden-record scorer	Relates datasets, finds joins, detects duplicate and overlapping datasets, recommends golden records, writes estate insights
Definition and lineage	Glossary drafter, refinement loop, conflict resolver, lineage inferrer, dictionary publisher	Definitions, conflict resolution, lineage, and publishing in every format
Reverse-SQL and BI	Report parser, metric mapper, join planner, SQL synthesizer, query checker, reconciler, BI extractor, BI script translator, BI expression translator	Reconstructs report SQL by translating BI logic or by inferring it from outputs, then validates it

8. Knowledge model and trust

Sources contain schemas, which contain datasets, which contain columns. Documents describe them; columns join, match and derive from each other; business terms define them; reports and BI applications compute metrics from them. Every fact records **when it was true in the data** and **when the system believed it**, so any past state can be reconstructed.

Trust level	Source	Example
1. Confirmed	Human decision	"order_id + line_no is the key"
2. Documented	Documents, schemas, view definitions, BI scripts	Dictionary entry, view SQL, Qlik load statement
3. Validated	Agent finding checked against the full data	Key proven unique across 1.2M rows
4. Inferred	Agent reasoning, not yet checked	Meaning guessed from names and statistics

When levels disagree (a dictionary says a column is unique, but the data has duplicates), a conflict is raised for review instead of overwriting either fact. Confirmed facts are never changed, only superseded by a newer confirmation.

9. Working with the data estate

Dremio is the primary source of data and metadata. The system reads it read-only and pushes heavy computation (statistics, uniqueness and containment checks) down to Dremio, so analysis scales with the estate rather than with our servers.

- **Knowledge the estate already holds.** View definitions become lineage; joins in views and in query history become join evidence; wikis and tags become documented context.
- **Overlapping and duplicate datasets.** A layered check finds identical views, views with the same logic, subsets, snapshots and copies: definitions and lineage first (free), then schemas, then bounded data fingerprints only where needed. Datasets are grouped into clusters.
- **Golden records.** Each cluster gets a recommended golden record based on lineage position, completeness, freshness, usage, location and documentation. A data owner always confirms it.
- **Work per cluster, not per copy.** Matching, definitions and the dictionary work on the golden or best member and apply to its copies, which cuts review effort sharply.
- **Estate report.** Duplicate clusters, recommended golden records, unused datasets and reports that read a non-golden copy. Recommendations only; the estate is never changed.
- **Usage first.** Analysis starts with what reports and users actually read, then expands.

10. Reconstructing reports

Mode	When	How
Translate	The BI logic is available (Qlik scripts, project folders, QVD files)	BI logic is extracted into a tool-neutral model and translated into Dremio SQL, then checked and executed in Dremio
Infer	Only the report output is available (PDF, Excel)	Measures and dimensions are mapped to columns using the knowledge graph; candidate SQL is generated and refined until it reproduces the report across several periods
Hybrid	An exported report belongs to a known BI application	Translated, and the export is used as additional validation

Every reconstruction carries a validation level, from "parsed" and "executes" through "row counts match" and "data matches" (against reload logs and QVD files) to "figures match" (against exported reports) and, when server access exists, "matches the live BI engine". A known pitfall, double counting caused by how BI tools and SQL aggregate across joined tables, is checked explicitly. The first BI adapter covers QlikView and Qlik Sense; the model is tool-neutral so other BI tools can be added without changing the rest of the system.

11. Human in the loop

Mode	When	Example
Seed	Before a run	"cust_ref in this source is the customer number"
Ask	An agent cannot proceed	"Which value does variable vYear take?"
Gate	Before a high-impact fact is trusted	Confirming a key, a golden record or reconstructed SQL
Queue	Non-blocking batch review	Approving 30 column matches at once
Correct	A reviewer edits an output	Rewriting a definition; the edit becomes confirmed

Validated, high-confidence, low-impact facts are accepted automatically and stay reversible; uncertain or high-impact ones are queued or gated. The queue is ordered by impact and uncertainty, so review time

goes where it matters. Corrections are stored with the original proposal and reused as examples, so agents improve on the organization's data without retraining.

12. Technology

Layer	Choice
Workbench	Next.js with React Flow
API and identity	FastAPI; Keycloak single sign-on, federating to the client's identity provider
Orchestration	LangGraph with a Postgres checkpointer
Agent-to-knowledge tools	Model Context Protocol server (FastMCP)
AI models	Claude models through Google Cloud Vertex AI (Gemini available through the same gateway)
Knowledge store	Postgres fact log (system of record) with a Neo4j graph projection
Data access	Dremio via Arrow Flight SQL and its catalog API; DuckDB and Polars for files
SQL and BI	sqlglot (Dremio dialect); Qlik script grammar; QVD and reload-log readers
Matching	MinHash sketches, fuzzy matching, statistical containment checks
Privacy	Microsoft Presidio masking before any model call that includes samples
Observability	Langfuse and OpenTelemetry
Exports	OpenLineage, dbt YAML, Excel, Markdown, PDF
Deployment	Docker Compose locally; Kubernetes with Helm for staging, production and client installs

13. Security, privacy and deployment

- **Data minimization.** Only schemas, statistics, definitions and masked samples reach an AI model; raw data stays in the client environment. In the starting stage, no row-level data reaches a model.
- **Read-only access** to Dremio and BI applications through dedicated service accounts, with controlled query load.
- **Isolation.** Each client's knowledge is isolated in the store itself, and isolation fails closed; knowledge learned for one client is never used for another.
- **Access control.** Single sign-on with roles for viewers, reviewers, data owners and administrators.
- **Auditability.** An append-only record of every proposal, decision, prompt version and model version; any past state can be reconstructed.
- **Environments.** Local development first, then staging on Google Cloud, then production. The same build can run hosted by us in a dedicated project per client, or inside the client's own cloud or data centre with their approved model endpoint, including air-gapped installations.
- **Regulation.** Designed to support GDPR, India's DPDP Act and sector rules; SOC 2 or ISO 27001 certification planned before production sales.

14. Roadmap

Stage	Weeks	Focus
Starting stage: Qlik to SQL	1–12	Qlik scripts reconstructed as validated Dremio SQL, lineage across QVD chains, migration packs (see the starting-stage proposal)
Platform: profiling and estate	6–8	Dataset profiling in Dremio, sensitive-data tagging, keys; first duplicate and overlap detection; usage tracking
Platform: relationships and dictionary	9–12	Joins and matches across datasets; golden-record recommendations with owner confirmation; first data dictionary; production-shaped installation
Platform: inference and refinement	13–15	Reconstruction from report outputs alone; consistency checks, definition refinement and conflict resolution; first estate report
Platform: complete and evaluate	16–18	All exports, external access for other tools, cost dashboards, full evaluation, security review and load testing
Pilot	Following	Paid pilot with a client on a report family; live Qlik engine validation where server access is granted
Product v1 and scale	Next 6–12 months	Multi-client hosting, catalog connectors, additional BI tools, natural-language questions over the knowledge graph, impact analysis

Each stage ends with a review against measurable criteria. The go-to-market position: *"We reconstruct how your numbers are produced when documentation and code are missing, and we feed that knowledge into the catalog you already own."*

15. Success measures

Area	Measure	Target
Keys	Primary-key accuracy (F1) on the test set	At least 90%
Joins	Foreign-key accuracy (F1) within a source	At least 80%
Estate	Precision of identical / same-logic dataset detection	At least 95%
Estate	Recommended golden records accepted by owners	At least 80%
Reconstruction	BI script statements translated	At least 90%
Reconstruction	Report figures reproduced where evidence exists (standard constructs)	At least 70%
Trust	Automatically accepted facts later reversed by reviewers	Below 5%
Compounding	Review items per new dataset from the same source	Declining over time
Freshness	Late-arriving documents update affected facts without manual re-runs	Always
Cost	Cost per dataset and per report	Measured and reported

16. Risks and mitigations

Risk	Mitigation
Wrong facts spread through the knowledge graph	Trust levels, validation against data, review gates, full history and rollback; confirmed facts never change
Duplicate datasets cause repeated work or facts attached to the wrong copy	Equivalence clusters; work on the golden or best member; owners confirm golden records

Risk	Mitigation
Reconstructed SQL matches by coincidence	Validation across several periods, slices and independent evidence; validation levels reported
AI output varies between runs	Structured outputs, pinned prompt and model versions, response memoization, regression evaluation before any model change
Load on the client's Dremio	Bounded and cached queries, concurrency limits, off-peak scheduling, visible query labels
Reviewer fatigue	Impact-ranked queue, batching, per-cluster review, automatic acceptance of validated low-impact facts
Cost overruns	Per-task and per-run budgets, caching, model routing by task difficulty
Overlap with incumbent catalogs	Positioned as an inference engine that feeds catalogs

17. Decision requested

Approve the Agentic Dataset Context Engine programme as described: the 12-week Qlik-to-SQL starting stage, followed by the platform stages through week 18, with read-only access to Dremio and the Qlik artifacts, a Google Cloud project for model access, and an infrastructure and model budget to be sized on Google Cloud pricing in the first two weeks. Progress is reviewed at the end of every stage against section 15.

18. References

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